

YOUTUBE CHANNEL: S KHAN ACADEMY

MTH643 CURRENT PAPER FINAL FILE

SOLVED BY SANAULLAH KHAN

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Subject Enrolment

1. Online classes are available (Mathematics)
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 - Assignment's
 - Quizzes
 - GDB's
3. Solved quiz files
4. Past papers
5. Any info about virtual university of Pakistan

Final project (MTH620)

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S KHAN ACADEMY

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Question

Find sum and write program $\sum_{k=1}^n \left(\frac{3}{x}\right)^k$ and take the value as an input from user then run the program by giving n as 5

CODE:

```
clc
clear all
close all
syms x
n=input('Enter the value of n: ');
sum=0;
for k=1:n
    sum=sum+(3/x)^k;
end
disp(sum)
```

Question

Find the numerical solution of the following equation using ode45 command and plot the curve on [-2 2].

$$y'(x) = y - 3x^2 + 7 \quad y(0) = 1$$

CODE:

```
clc
clear all
close all
f=@(x,y)y-3*x^2+7;
interval=[-2,2];
y0=1;
[x,y]=ode45(f,interval,y0)
plot(x,y)
```

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Question

$\int_1^2 (\cos(x) - \log x + e^x) dx$ solve the equation using Simpson's 1/8 and keep 10 sub intervals.

CODE:

```
clc
clear all
close all
f=@(x)(cos(x)-log(x)+exp(x));
a=1;
b=2;
n=10;
h=(b-a)/n;
sumofodd=0;
sumofeven=0;
for k=1:n-1
    x(k)=a+k*h;
    if mod(k,2)==0
        sumofeven=sumofeven+f(x(k));
    else
        sumofodd=sumofodd+f(x(k));
    end
end
answer=h/3*(f(a)+f(b)+2*sumofeven+4*sumofodd)
```

Question

Find the numerical solution of the following equation using ode45 command and plot the curve on [0 2].

$$y'(x) = y - x^2 + 1 \quad y(0) = 0.5$$

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CODE:

```
clc
clear all
close all
f=@(x,y)y-x^2+1;
interval=[0,2];
y0=0.5;
[x,y]=ode45(f,interval,y0)
plot(x,y)
```

Question

Find the numerical solution of the following equation using ode45 command and plot the curve on [0 2].

$$y'(x) = y - x^2 + 1 \quad y(0) = 0.5:0.3:3$$

CODE:

```
clc
clear all
close all
f=@(x,y)y-x^2+1;
interval=[0,2];
y0=0.5:0.3:3;
[x,y]=ode45(f,interval,y0)
plot(x,y)
```

Question

Find the numerical solution of the following equation using ode45 command and plot the curve on x=0:0.2:2

$$y'(x) = y - x^2 + 1 \quad y(0) = 0.5$$

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CODE:

```
clc
clear all
close all
f=@(x,y)y-x^2+1;
interval=0:0.2:2;
y0=0.5;
[x,y]=ode45(f,interval,y0)
plot(x,y)
```

Question

Write a for loop to add first 20 numbers which are divisible by 5.

CODE:

```
clc
clear all
close all
sum=0;
for i=5:5:100;
sum=sum+i;
end
disp(sum)
```

Question

Write a program which takes a value from user and then using for loop display all of its divisors.

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CODE:

```
clc
clear all
close all
x=input('Enter a number:')
k=x;
for i=1:k;
if mod(x,i)==0
disp(i)
end
end
```

Question

Numerically approximate the solution of the following ode $dy/dt=2t$ use time interval [0 5] and initial condition is $y(0)=0$ by using ode45 command.

CODE:

```
clc
clear all
close all
f=@(t,y)2*t;
interval=[0 5];
y0=0;
[t,y]=ode45(f,interval,y0)
plot(t,y)
```

Question

Write a function with name 'myfunction' which takes any two number as input and returns the greater number.

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CODE:

```
function[x,y]=myfunction(a,b)
if a>b
disp('The max number is a')
else
disp('The max number is b')
end
```

Question

Write a function with name 'newfunction' which takes any two number as input and returns the highest common factor of the number.

CODE:

```
function[HCF]=newfunction(a,b)
for k=1:1:(a+b);
if mod(a,k)==0 && mod(b,k)==0
HCF=k;
end
end
```

Question

Write a program which prints all the numbers divisible by 3 and 5 for a given number.

CODE:

```
clc
clear all
close all
x=input('Enter The number:')
n=x;
for k=1:n;
if mod(k,3)==0 && mod(k,5)==0
disp(k)
end
end
```

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13 Question

Write a for loop which adds first 30 odd numbers.

CODE:

```
clear all
close all
clc
i=1:60;
sum=0;
for k=1:2:length(i);
sum=sum+k;
end
disp(sum)
```

14 Question

Write a function name cube takes a number as an input if the number is even return
Cube of that number if odd then return square.

CODE:

```
function[cube square]=cube(n)
if rem(n,2)==1
square=n^2
else
cube=n^3
end
end
```

15 Question

Find sum and write program $\sum_{k=1}^n \left(\frac{3}{x}\right)^k$ and input x=4 and n=12.

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CODE:

```
clc
clear all
close all
n=12;
x=4;
sum=0;
for k=1:n
sum=sum+(3/x)^k;
end
disp(sum)
```

Question

Solve by using the newton Raphson's method $x^3 + x$ by using for loop when $x_0=2$ and $n=3$.

CODE:

```
clc
clear all
close all
f=@(x)x^3+x
df=@(x)3*x^2+1
x0=2;
n=3;
x=x0;
for i=1:n;
x=x-f(x)/df(x)
end
```

Question

Solve by using the newton Raphson's method $x^3 + x$ by using for loop when $x_0=2$ and $n=3$.

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CODE:

```
clc
clear all
close all
f=@(x)x^3+x
df=@(x)3*x^2+1
x0=2;
n=3;
x=x0;
tol=10^-4
while abs(f(x))>tol
x=x-f(x)/df(x)
end
fprintf('Root: %0.15 f\n',x)
```

Question

Write a function with the name 'practisefunction' which takes any three random variables as an input and returns the number in ascending order.

CODE:

```
function [Ascending]=practisefunction(a,b,c)
A=[a,b,c]
Ascendingorder=sort(A)
end
```

Question

Find factorial of a number by using while loop.

CODE:

```
clc
clear all
close all
```

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```
n=input('enter a number:');
i=n;
fact=1;
if i==0
fact=1;
else if i<0
disp('the number you are entered is nagative')
end
end
while i>0
fact=fact*i;
i=i-1;
end
disp(fact)
```

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Question

Find factorial of a number by using for loop.

CODE:

```
clc
clear all
close all
n=input('Enter a number:');
fact=1;
for i=1:n;
fact=fact*i;
end
disp(fact)
```

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Question

write a function with name examquestions which take any two number as an input and return the sum and product of those number as output.

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CODE:

```
function[a b]=examquestions(a,b)
sum=a+b
product=a*b
end
```

Question

Use while loop to calculate the factorial of 10.

CODE:

```
clc
clear all
close all
n=10;
i=n;
fact=1;
while i>0
fact=fact*i;
i=i-1;
end
disp(fact)
```

Question

Numerically solve the following equation by using ode45 command

$$\frac{dy}{dx} = \frac{y}{t^2} + \frac{2}{t} \quad y(0) = 1 \quad [2, -2]$$

CODE:

```
clc
clear all
close all
f=@(t,y)(y/t^2+2/t);
interval=[-2 2];
y0=1;
[t,y]=ode45(f,interval,y0)
```

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plot(t,y)

Question

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 11 & 5 & 7 \\ 2 & 4 & 6 \\ 3 & 5 & 5 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} 1 & 8 & 4 \\ 3 & 4 & 7 \\ 2 & 7 & 9 \\ 9 & 3 & 6 \\ 6 & 7 & 5 \\ 4 & 4 & 7 \end{bmatrix}$$

Find

1. Inv(A)
2. A*B

CODE:

```
clc
clear all
close all
A=sym([3/11 2/5 4/7;2/3 4/5 6/5]);
B=sym([1/3 8/4 4;2/9 3/7 9/6;6 7/4 5/7]);
C=inv(B)
D=sym(A*B)
```

Question

$$2x + y + z = 2, \quad -x + y - z = 3, \quad x + 2y + 3z = -10$$

1. Write system in MATLAB as an augmented matrix
2. Make reduced echelon form

CODE:

```
clc
clear all
close all
A=[2,1,1;-1,1,-1;1,2,3]
b=[2;3;-10]
```

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Ab=[A b]

%AX=B%

C=rref(Ab)

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Question

$y(x) = \sqrt{1+x^3}$ solve the equation using Simpson's 3/8 rule interval [1 5] and keep 12 sub interval.

CODE:

clc

clear all

close all

f=@(x)sqrt(1+x^3);

a=1;

b=5;

n=12;

% if step size is given the we use n=(b-a)/h

% if subinterval is given the we use h=(b-a)/n

h=(b-a)/n;

sum1=0;

sum2=0;

for i=1:n-1;

x(i)=a+i*h;

if rem(i,3)==0

sum1=sum1+f(x(i));

else

sum2=sum2+f(x(i));

end

end

Area=3*h/8*(f(a)+f(b)+2*sum1+3*sum2)

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Question

$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 7$ and $y(0) = 1$ and $y'(0) = 0$

Solve the differential equation.

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CODE:

```
★clc
★clear all
★close all
★syms y(t)
★Dy=diff(y,t)
★ode=diff(y,t,2)+2*diff(y,t)+5*y==7
★cond1=y(0)==1
★cond2=Dy(0)==0
★cond=[cond1,cond2]
★sol=dsolve(ode,cond)
```

Question

Solve $x - 2\sin(x^2) = 0$ by using bisection method perform 10 iterations to find the root between 0.4 and 0.6.

CODE:

```
★clc
★clear all
★close all
★f=@(x)x-2*sin(x^2)
★a=0.4;
★b=0.6;
★n=10;
★if f(a)*f(b)>0
★disp('there is no change of sign:')
★return
★end
★tol=10^-4;
★disp('iteration number a b c')
★i=1;
★while (abs(a-b)>=tol)
★c=(a+b)/2;
```

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```
if f(c)==0;
fprintf('root of the function is %f:',c)
return
end
fprintf('%2i \t %f \t %f \t %f \n',i,a,b,c)
if f(a)*f(c)>0
a=c;
else
b=c;
i=i+1;
end
end
m=(a+b)/2
fprintf('root lies at c=%f \n',m)
```

Question

Write a function with name 'circlecalculation' which take the circle of radius as input and circf as output.

CODE:

```
function [circf]=circlecalculation(radius)
radius=input('Enter the radius of circle:')
circf=2*pi*radius
end
```

Question

$y(x) = \sqrt{1+x^3}$ solve the equation using Simpson's 1/8 rule interval [1 5] and keep 8 sub interval.

CODE:

```
clc
clear all
close all
f=@(x)sqrt(1+x^3)
a=1;
```

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```
b=5;
n=8;
% if step size is given the we use n=(b-a)/h
% if subinterval is given the we use h=(b-a)/n
h=(b-a)/n;
sumofeven=0;
sumofodd=0;
for k=1:n-1;
    x(k)=a+k*h;
    if rem(k,2)==0
        sumofeven=sumofeven+f(x(k));
    else
        sumofodd=sumofodd+f(x(k));
    end
end
Area=h/3*(f(a)+f(b)+2*sumofeven+4*sumofodd)
```

Question

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 11 & 5 & 7 \\ 2 & 4 & 6 \\ 3 & 5 & 5 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} 1 & 8 & 4 \\ 3 & 4 & 7 \\ 2 & 7 & 9 \\ 9 & 3 & 6 \\ 6 & 7 & 5 \\ 4 & 4 & 7 \end{bmatrix}$$

Find

1. Sum of 3rd row of B
2. Sum of 2nd column of A

CODE:

```
clc
clear all
close all
A=sym([3/11 2/5 4/7;2/3 4/5 6/5])
B=sym([1/3 8/4 4;2/9 3/7 9/6;6 7/4 5/7])
SumA=sum(A(:,2))
```

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```
SumB=sum(B(3,:))
```

Question

Solve the following equation $\frac{x^2}{2}$ by using trapezoidal rule in the interval [4 6] by using 10 sub interval.

CODE:

```
clc
clear all
close all
f=@(x)x^2/2
a=4;
b=6;
n=10;
% if step size is given the we use n=(b-a)/h
% if subinterval is given the we use h=(b-a)/n
h=(b-a)/n;
sum=0;
for i=1:n-1;
    x(i)=a+i*h;
    y(i)=f(x(i));
    sum=sum+y(i);
end
Area=h/2*(f(a)+f(b)+2*sum)
```

Question

Solve the following equation $\frac{x^2}{2}$ by using Reimann sum in the interval [4 6] by using 10 sub interval.

CODE:

```
clc
clear all
```

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```
close all
f=@(x)x^2/2
a=4;
b=6;
n=10;
h=(b-a)/n;
value=0;
for i=1:n;
c=a+i*h;
value=value+f(c);
end
Area=value*h
```

Question

Find the value of X if

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

CODE:

```
clc
clear all
close all
A=[1,3;2,4]
B=[-2;3]
% As we know that
% AX=B
% X=inv(A)*B
X=inv(A)*B
```

Question

Solve the system of linear equation by using solve command.

$$3x + 2y + x = 1, \quad y = 2, \quad x + 2y = 3$$

CODE:

```
clc
```

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```
clear all
close all
eqn1='3*x+2*y+z=1';
eqn2='y=2';
eqn3='x+2*y=3';
sol=solve(eqn1,eqn2,eqn3)
Answer=double([sol.x;sol.y;sol.z])
```

Question

Find the eigen values and eigen vector of the following matrix.

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 7 & 8 & 9 \\ 3 & 1 & 3 \end{bmatrix}$$

CODE:

```
clc
clear all
close all
A=[3,1,4;7,8,9;3,1,3]
V=eig(A)
[P D]=eig(A)
```

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